UNIT 3

Classification of Elements and Periodicity in Properties

The Periodic Table is arguably the most important concept in chemistry, both in principle and in practice. It is the everyday support for students, it suggests new avenues of research to professionals, and it provides a succinct organization of the whole of chemistry.

What this unit sets out to do

- appreciate how the concept of grouping elements in accordance to their properties led to the development of the Periodic Table, and understand the Periodic Law
- understand the significance of atomic number and electronic configuration as the basis for periodic classification, and name the elements with Z greater than 100 according to IUPAC nomenclature
- classify elements into s, p, d and f blocks and learn their main characteristics, recognise the periodic trends in physical and chemical properties, and compare the reactivity of elements and correlate it with their occurrence in nature
- explain the relationship between ionization enthalpy and metallic character, and use scientific vocabulary appropriately to communicate ideas about atomic and ionic radii, ionization enthalpy, electron gain enthalpy, electronegativity and valence

3.1 Why do we need to classify elements

- the Periodic Table is a remarkable demonstration of the fact that the chemical elements are not a random cluster of entities but instead display trends and lie together in families, and an awareness of it is essential to anyone who wishes to disentangle the world and see how it is built up from the chemical elements
- the elements are the basic units of all types of matter, and in 1800 only 31 elements were known, by 1865 the number of identified elements had more than doubled to 63, and at present 114 elements are known, of which the recently discovered ones are man-made, while efforts to synthesise new elements are continuing
- with such a large number of elements it is very difficult to study individually the chemistry of all of them and their innumerable compounds, so to ease out this problem scientists searched for a systematic way to organise their knowledge by classifying the elements, which would not only rationalize known chemical facts but even predict new ones

In this unit

3.2 Genesis of Periodic Classification

3.3 Modern Periodic Law

3.4 Nomenclature of Elements

3.5 Electronic Configurations and the Table

3.6 Electronic Configurations

3.7 Periodic Trends in Properties

3.2 Genesis of periodic classification

Dobereiner's triads

- the classification of elements into groups and the development of the Periodic Law and Periodic Table are the consequences of systematising the knowledge gained by a number of scientists through their observations and experiments
- the German chemist Johann Dobereiner in the early 1800s was the first to consider the idea of trends among properties of elements, and by 1829 he noted a similarity among the physical and chemical properties of several groups of three elements, called triads
- in each case he noticed that the middle element of a triad had an atomic weight about half way between the atomic weights of the other two, and that its properties were also in between those of the other two members

Newlands' law of octaves

- the English chemist John Alexander Newlands in 1865 profounded the Law of Octaves, arranging the elements in increasing order of their atomic weights and noting that every eighth element had properties similar to the first
- the relationship was just like every eighth note that resembles the first in octaves of music, and his Law of Octaves seemed to be true only for elements up to calcium
- although his idea was not widely accepted at that time, he was later awarded the Davy Medal in 1887 by the Royal Society, London, for his work
- the Periodic Law as we know it today owes its development to the Russian chemist Dmitri Mendeleev, 1834-1907, and the German chemist Lothar Meyer, 1830-1895

Table 3.A Dobereiner's Triads

Element	Atomic weight	Element	Atomic weight	Element	Atomic weight
Li	7	Ca	40	Cl	35.5
Na	23	Sr	88	Br	80
K	39	Ba	137	I	127

Table 3.B Newlands' Octaves

Element	Li	Be	B	C	N	O	F
At. wt.	7	9	11	12	14	16	19
Element	Na	Mg	Al	Si	P	S	Cl
At. wt.	23	24	27	29	31	32	35.5
Element	K	Ca					
At. wt.	39	40					

Lothar Meyer, and the law itself

- unlike Newlands, Lothar Meyer observed a change in the length of the repeating pattern, and by 1868 he had developed a table of the elements closely resembling the Modern Periodic Table, however his work was not published until after that of Dmitri Mendeleev, the scientist generally credited with the development of the Modern Periodic Table
- while Dobereiner initiated the study of periodic relationship, it was Mendeleev who was responsible for publishing the Periodic Law for the first time, stating that the properties of the elements are a periodic function of their atomic weights
- he arranged elements in horizontal rows and vertical columns in order of increasing atomic weight in such a way that the elements with similar properties occupied the same vertical column or group

Where he broke his own order, and where he left gaps

- his system was more elaborate than Lothar Meyer's, since he fully recognized the significance of periodicity and used a broader range of physical and chemical properties, relying in particular on similarities in the empirical formulas and properties of the compounds the elements form
- he realized that some elements did not fit if the order of atomic weight was strictly followed, so he ignored that order, thinking the atomic measurements might be incorrect, and placed elements with similar properties together, for example iodine, with a lower atomic weight than tellurium, was placed alongside fluorine, chlorine and bromine
- keeping his primary aim of arranging elements of similar properties in the same group, he proposed that some elements were still undiscovered and left several gaps, for example both gallium and germanium were unknown when he published, so he left a gap under aluminium and one under silicon and called them Eka-Aluminium and Eka-Silicon
- he predicted not only the existence of gallium and germanium but also described some of their general physical properties, and the boldness of these quantitative predictions and their eventual success made him and his Periodic Table famous

Table 3.C Mendeleev's Predictions for the Elements Eka-aluminium (Gallium) and Eka-silicon (Germanium)

Property	Eka-aluminium (predicted)	Gallium (found)	Eka-silicon (predicted)	Germanium (found)
Atomic weight	68	70	72	72.6
Density/(g/cm3)	5.9	5.94	5.5	5.36
Melting point/K	Low	302.93	High	1231
Formula of oxide	E2O3	Ga2O3	EO2	GeO2
Formula of chloride	E Cl3	GaCl3	ECl4	GeCl4

Elements set out in groups running across and series running down

Each group headed by the general formula of the oxide and the hydride its members form

Gaps left open where he expected an element that had not yet been found

Figure 3.A Mendeleev's Periodic Table published earlier

What this page carries

- the whole of this page is given over to Mendeleev's Periodic Table as it was published, headed the periodic system of the elements in groups and series
- the table is printed as a drawing rather than as text, so the arrangement itself is in the book and is described here rather than reproduced

3.3 Modern periodic law and the present form of the periodic table

Moseley, and what he showed

- when Mendeleev developed his Periodic Table chemists knew nothing about the internal structure of the atom, however the beginning of the 20th century witnessed profound developments in theories about sub-atomic particles
- in 1913 the English physicist Henry Moseley observed regularities in the characteristic X-ray spectra of the elements, and a plot of the square root of the frequency of the X-rays emitted against atomic number gave a straight line where the same plot against atomic mass did not, so he showed that the atomic number is a more fundamental property of an element than its atomic mass
- Mendeleev's Periodic Law was accordingly modified into what is known as the Modern Periodic Law, that the physical and chemical properties of the elements are periodic functions of their atomic numbers
- the Periodic Law revealed important analogies among the 94 naturally occurring elements, neptunium and plutonium being found in pitch blende, an ore of uranium, like actinium and protoactinium, and it stimulated renewed interest in Inorganic Chemistry and has carried into the present with the creation of artificially produced short-lived elements

Periods and groups in the long form

- the atomic number is equal to the nuclear charge, that is the number of protons, or the number of electrons in a neutral atom, so it is easy to visualize the significance of quantum numbers and electronic configurations in the periodicity of elements, and it is now recognized that the Periodic Law is essentially the consequence of the periodic variation in electronic configurations
- numerous forms of Periodic Table have been devised, some emphasising chemical reactions and valence and others the electronic configuration, and the modern long form is the most convenient and widely used, its horizontal rows, which Mendeleev called series, being called periods and its vertical columns groups
- elements having similar outer electronic configurations are arranged in vertical columns referred to as groups or families, and on the recommendation of the International Union of Pure and Applied Chemistry the groups are numbered from 1 to 18, replacing the older notation of IA to VIIA, VIII, IB to VIIB and 0
- there are altogether seven periods, the period number corresponding to the highest principal quantum number of the elements in it, the first containing 2 elements and the subsequent ones 8, 8, 18, 18 and 32, while the seventh is incomplete and like the sixth would have a theoretical maximum of 32, and 14 elements of both the sixth and seventh periods, the lanthanoids and actinoids, are placed in separate panels at the bottom

3.4 Nomenclature of elements with atomic numbers greater than 100

Who names a new element

- the naming of new elements had traditionally been the privilege of the discoverer, with the suggested name ratified by the IUPAC, and in recent years this has led to some controversy
- the new elements with very high atomic numbers are so unstable that only minute quantities, sometimes only a few atoms of them, are obtained
- Glenn T. Seaborg's work in the middle of the 20th century, starting with the discovery of plutonium in 1940 and followed by all the transuranium elements from 94 to 102, led to the reconfiguration of the periodic table placing the actinoids below the lanthanoids, and in 1951 he was awarded the Nobel Prize in chemistry for his work, element 106 being named Seaborgium in his honour

Every element in its place, with atomic number and ground state outer electronic configuration under each

Groups numbered 1 to 18 across the top

Lanthanoids and actinoids lifted out into two panels below the main body

Figure 3.B Long form of the Periodic Table of the Elements with their atomic numbers and ground state outer electronic configurations. The groups are numbered 1-18 in accordance with the 1984 IUPAC recommendations. This notation replaces the old numbering scheme of IA–VIIA, VIII, IB–VIIB and 0 for the elements.

What this page carries

- the whole of this page is given over to the long form of the Periodic Table, the version this unit works from
- the old group labels IA to VIIA, VIII, IB to VIIB and 0 are still printed along the top for comparison with the numbering that replaced them
- the table is printed as a drawing rather than as text, so it is described here rather than reproduced

Why a systematic name was needed

- only minute quantities of the very heavy elements are obtained, sometimes only a few atoms, so their synthesis and characterisation require highly sophisticated costly equipment and laboratories, and such work is carried out with competitive spirit in only some laboratories in the world
- scientists, before collecting reliable data on a new element, at times get tempted to claim its discovery, for example both American and Soviet scientists claimed credit for discovering element 104, the Americans naming it Rutherfordium and the Soviets Kurchatovium
- to avoid such problems the IUPAC has recommended that until a new element's discovery is proved and its name officially recognised, a systematic name be derived directly from the atomic number using numerical roots for 0 and the numbers 1 to 9, the roots being put together in the order of the digits that make up the atomic number with ium added at the end

Table 3.D Notation for IUPAC Nomenclature of Elements

Digit	Name	Abbreviation	Digit	Name	Abbreviation
0	nil	n	5	pent	p
1	un	u	6	hex	h
2	bi	b	7	sept	s
3	tri	t	8	oct	o
4	quad	q	9	enn	e

Table 3.E Nomenclature of Elements with Atomic Number Above 100

Atomic Number	Name according to IUPAC nomenclature	Symbol	IUPAC Official Name	IUPAC Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Ununnillium	Uun	Darmstadtium	Ds
111	Unununnium	Uuu	Rontgenium	Rg
112	Ununbium	Uub	Copernicium	Cn
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

Temporary name, then permanent one

- the new element first gets a temporary name, with a symbol consisting of three letters, and later a permanent name and symbol are given by a vote of IUPAC representatives from each country
- the permanent name might reflect the country, or the state of the country, in which the element was discovered, or pay tribute to a notable scientist
- as of now, elements with atomic numbers up to 118 have been discovered, and official names of all elements have been announced by IUPAC

Problem 3.1

- the question asks for the IUPAC name and symbol for the element with atomic number 120
- the roots for 1, 2 and 0 are un, bi and nil respectively, so the symbol is Ubn and the name is unbinilium

3.5 Electronic configurations of elements and the periodic table

Where an element sits, and why

- an electron in an atom is characterised by a set of four quantum numbers, the principal one defining the main energy level known as the shell, and the distribution of electrons into the orbitals of an atom is called its electronic configuration
- an element's location in the Periodic Table reflects the quantum numbers of the last orbital filled, so there is a direct connection between the electronic configurations of the elements and the long form of the table
- the period indicates the value of n for the outermost or valence shell, so each successive period is associated with the filling of the next higher principal energy level, and the number of elements in each period is twice the number of atomic orbitals available in the level being filled
- the first period starts with the filling of the lowest level, 1s, and therefore has two elements, hydrogen and helium, when the first shell is completed, while the second period starts with lithium whose third electron enters the 2s orbital, then beryllium with four electrons at 1s2 2s2, and from boron onwards the 2p orbitals begin to fill

Where the inner transition series sit, and Problem 3.2

- actinium at Z = 89 gives the 5f-inner transition series known as the actinoid series, and the 4f- and 5f-inner transition series are placed separately in the Periodic Table to maintain its structure and to preserve the principle of keeping elements with similar properties in a single column
- Problem 3.2 asks how the presence of 18 elements in the 5th period is justified, and when n = 5 the values of l are 0, 1, 2 and 3, the order in which the energy of the available 4d, 5s and 5p orbitals increases being 5s below 4d below 5p, giving 9 orbitals in all, which can accommodate at most 18 electrons, so there are 18 elements in the 5th period

Groupwise electronic configurations

- elements in the same vertical column or group have similar valence shell electronic configurations, the same number of electrons in the outer orbitals, and similar properties
- the Group 1 elements, the alkali metals, all have an ns1 valence shell configuration, running lithium at helium core plus 2s1, sodium at neon core plus 3s1, potassium at argon core plus 4s1, rubidium at krypton core plus 5s1, caesium at xenon core plus 6s1 and francium at radon core plus 7s1
- so the properties of an element have a periodic dependence upon its atomic number and not on relative atomic mass

3.6 Electronic configurations and types of elements, the s, p, d and f blocks

The four blocks, and two exceptions

- the aufbau principle and the electronic configuration of atoms provide a theoretical foundation for the periodic classification, the elements in a vertical column constituting a group or family and exhibiting similar chemical behaviour because they have the same number and same distribution of electrons in their outermost orbitals
- we can classify the elements into four blocks, s-block, p-block, d-block and f-block, depending on the type of atomic orbitals being filled with electrons
- there are two exceptions, helium, which strictly belongs to the s-block but is positioned in the p-block along with the other group 18 elements because it has a completely filled valence shell and so exhibits properties characteristic of the other noble gases, and hydrogen, which has only one s-electron and so can be placed in group 1 with the alkali metals, but can also gain an electron to reach a noble gas arrangement and behave like a group 17 halogen, so being a special case it is placed separately at the top of the table

3.6.1 The s-block elements

Which elements they are

- the elements of Group 1, the alkali metals, and Group 2, the alkaline earth metals, which have ns1 and ns2 outermost electronic configuration, belong to the s-block elements

The table shaded by which kind of orbital is filling, s, p, d or f

The same table shaded a second way, into metals, non-metals and metalloids

Hydrogen set apart at the top, since it fits neither group 1 nor group 17 cleanly

Figure 3.C The types of elements in the Periodic Table based on the orbitals that are being filled. Also shown is the broad division of elements into METALS ( ), NON-METALS ( ) and METALLOIDS ( ).

What this page carries

- the whole of this page is given over to the Periodic Table shaded by block and by the broad division into metals, non-metals and metalloids
- the table is printed as a drawing rather than as text, so it is described here rather than reproduced

What the s-block elements are like

- they are all reactive metals with low ionization enthalpies, losing the outermost electrons readily to form a 1+ ion in the case of the alkali metals or a 2+ ion in the case of the alkaline earth metals
- the metallic character and the reactivity increase as we go down the group, and because of their high reactivity they are never found pure in nature
- the compounds of the s-block elements, with the exception of those of lithium and beryllium, are predominantly ionic

Transition elements, and the three that are not quite

- the transition elements are marked by variable valence, that is variable oxidation states, by paramagnetism, and by being often used as catalysts
- Zn, Cd and Hg, which have the configuration n minus 1 d10 ns2, do not show most of the properties of the transition elements
- in a way the transition metals form a bridge between the chemically active metals of the s-block and the less active elements of Groups 13 and 14, and so take their familiar name transition elements

3.6.2 The p-block elements

Groups 13 to 18

- the p-block elements comprise those belonging to Group 13 to 18, and these together with the s-block elements are called the representative elements or main group elements, the outermost configuration varying from ns2 np1 to ns2 np6 in each period
- at the end of each period is a noble gas element with a closed valence shell ns2 np6 configuration, all of whose valence orbitals are completely filled, and it is very difficult to alter this stable arrangement by the addition or removal of electrons, so the noble gases exhibit very low chemical reactivity
- preceding the noble gas family are two chemically important groups of non-metals, the halogens in Group 17 and the chalcogens in Group 16, which have highly negative electron gain enthalpies and readily add one or two electrons respectively to attain the stable noble gas configuration
- the non-metallic character increases as we move from left to right across a period and the metallic character increases as we go down the group

3.6.3 The d-block elements, and 3.6.4 the f-block elements

The two inner blocks

- the d-block elements are those of Group 3 to 12 in the centre of the Periodic Table, characterised by the filling of inner d orbitals, with the general outer configuration n minus 1 d1 to d10 ns0 to ns2, except for Pd whose configuration is 4d10 5s0, and they are all metals which mostly form coloured ions
- the two rows of elements at the bottom of the table, the lanthanoids from cerium at Z = 58 to lutetium at Z = 71 and the actinoids from thorium at Z = 90 to lawrencium at Z = 103, are characterised by the outer configuration n minus 2 f1 to f14, n minus 1 d0 to d1, ns2
- the last electron added to each of these is filled into an f orbital, which is why the two series are called the inner-transition elements or f-block elements, and they are all metals whose properties within each series are quite similar

3.6.5 Metals, non-metals and metalloids

The two broad kinds, and the band between them

- the elements can be divided into metals and non-metals, metals comprising more than 78 per cent of all known elements and appearing on the left side of the Periodic Table, usually solids at room temperature, mercury being an exception while gallium and caesium have very low melting points at 303 K and 302 K, with high melting and boiling points, good conduction of heat and electricity, malleable so they can be flattened into thin sheets by hammering, and ductile so they can be drawn into wires
- non-metals are located at the top right hand side of the table, so in a horizontal row the property of elements changes from metallic on the left to non-metallic on the right, and non-metals are usually solids or gases at room temperature with low melting and boiling points, boron and carbon being exceptions, are poor conductors of heat and electricity, and most non-metallic solids are brittle and neither malleable nor ductile
- the elements become more metallic as we go down a group and the non-metallic character increases from left to right across the table, and the change is not abrupt but is marked by a thick zig-zag line, the elements bordering it and running diagonally across the table, such as silicon, germanium, arsenic, antimony and tellurium, showing properties characteristic of both metals and non-metals, and being called semi-metals or metalloids

Problem 3.4

- the question asks, considering atomic number and position in the periodic table, to arrange Si, Be, Mg, Na and P in the increasing order of metallic character
- metallic character increases down a group and decreases along a period from left to right, so the order of increasing metallic character is P, then Si, then Be, then Mg, then Na

3.7 Periodic trends in properties of elements, and 3.7.1 physical properties

The patterns, and which ones this unit follows

- there are many observable patterns as we descend a group or move across a period, for example within a period chemical reactivity tends to be high in the Group 1 metals, lower towards the middle of the table, and increases to a maximum in the Group 17 non-metals, while within a group of representative metals such as the alkali metals reactivity increases down the group, whereas within a group of non-metals such as the halogens it decreases
- to answer why the properties follow these trends we must look into the theories of atomic structure and the properties of the atom, and explain them in terms of the number of electrons and the energy levels
- numerous physical properties show periodic variation, such as melting and boiling points, heats of fusion and vaporization and energy of atomization, however the trends discussed here are those in atomic and ionic radii, ionization enthalpy, electron gain enthalpy and electronegativity

Atomic radius, and why it is hard to pin down

- finding the size of an atom is a lot more complicated than measuring the radius of a ball, firstly because the size of an atom, about 1.2 Å in radius, is very small

Covalent radius and metallic radius

- secondly, since the electron cloud surrounding the atom does not have a sharp boundary, the determination of the atomic size cannot be precise, so there is no practical way by which the size of an individual atom can be measured, though an estimate can be made by knowing the distance between the atoms in the combined state
- one practical approach for a non-metallic element is to measure the distance between two atoms when they are bound together by a single bond in a covalent molecule, and from this the covalent radius of the element can be calculated, for example the bond distance in the chlorine molecule is 198 pm and half of it, 99 pm, is taken as the atomic radius of chlorine
- for metals we define the metallic radius, taken as half the internuclear distance separating the metal cores in the metallic crystal, and for simplicity this book uses atomic radius to refer to either, depending on whether the element is a non-metal or a metal, atomic radii being measurable by X-ray or other spectroscopic methods

The two trends, and why

- the atomic size generally decreases across a period, because within the period the outer electrons are in the same valence shell while the effective nuclear charge increases with atomic number, so the electrons are pulled in more strongly
- within a family or vertical column the atomic radius increases regularly with atomic number, because as we descend the principal quantum number increases and the valence electrons are farther from the nucleus, the inner energy levels being filled with electrons which shield the outer ones from the pull of the nucleus
- the atomic radii of the noble gases are not considered here, since being monoatomic their non-bonded values are very large, and their radii should be compared not with the covalent radii but with the van der Waals radii of other elements

Table 3.F Atomic Radii/pm Across the Periods

Atom (Period II)	Li	Be	B	C	N	O	F
Atomic radius	152	111	88	77	74	66	64
Atom (Period III)	Na	Mg	Al	Si	P	S	Cl
Atomic radius	186	160	143	117	110	104	99

Table 3.G Atomic Radii/pm Down a Family

Atom (Group I)	Atomic Radius	Atom (Group 17)	Atomic Radius
Li	152	F	64
Na	186	Cl	99
K	231	Br	114
Rb	244	I	133
Cs	262	At	140

Radius falling steadily from lithium to fluorine

Every one of those atoms fills the same valence shell

The pull inward grows as the nuclear charge rises

Figure 3.D Variation of atomic radius with atomic number across the second period

Two rising lines, one for the alkali metals and one for the halogens

Each step down a group adds a shell, so the radius climbs

The halogen line sits below the alkali line throughout

Figure 3.E Variation of atomic radius with atomic number for alkali metals and halogens

Ionic radius

- the removal of an electron from an atom results in a cation, whereas gain of an electron leads to an anion, and ionic radii can be estimated by measuring the distances between cations and anions in ionic crystals, generally exhibiting the same trend as the atomic radii
- a cation is smaller than its parent atom because it has fewer electrons while its nuclear charge remains the same, and an anion is larger than its parent atom because the addition of electrons increases the repulsion among them and decreases the effective nuclear charge, so the ionic radius of the fluoride ion is 136 pm whereas the atomic radius of fluorine is only 64 pm, while the atomic radius of sodium is 186 pm against an ionic radius of 95 pm for Na+
- atoms and ions containing the same number of electrons are called isoelectronic species, for example O2-, F-, Na+ and Mg2+ all have ten electrons, and their radii differ because of their different nuclear charges, the cation with the greater positive charge having the smaller radius and the anion with the greater negative charge the larger, since there the net repulsion of the electrons outweighs the nuclear charge and the ion expands

Problem 3.5, and ionization enthalpy

- the question asks which of Mg, Mg2+, Al and Al3+ will have the largest and the smallest size, and since atomic radii decrease across a period, cations are smaller than their parent atoms, and among isoelectronic species the one with the larger positive nuclear charge has the smaller radius, the largest species is Mg and the smallest is Al3+
- a quantitative measure of the tendency of an element to lose an electron is given by its ionization enthalpy, which represents the energy required to remove an electron from an isolated gaseous atom in its ground state

Ionization enthalpy plotted against atomic number, one point per element

Sharp peaks at the noble gases, which have closed shells

Deep troughs at the alkali metals, which give up an electron readily

Figure 3.F Variation of first ionization enthalpies (ΔiH) with atomic number for elements with Z = 1 to 60

Second-period values rising from lithium to neon, with dips at boron and at oxygen

Alkali metal values falling steadily down the group

Together they show the across-a-period rise and the down-a-group fall

Figure 3.G First ionization enthalpies (ΔiH) of elements of the second period as a function of atomic number (Z) and Fig. 3.6(b) ΔiH of alkali metals as a function of Z.

First and second ionization enthalpy

- the first ionization enthalpy for an element X is the enthalpy change for X gas going to X plus gas and an electron, and it is expressed in units of kJ mol-1
- the second ionization enthalpy is the energy required to remove the second most loosely bound electron, that is to take X plus gas to X two plus gas and an electron
- energy is always required to remove electrons from an atom, so ionization enthalpies are always positive, and the second will be higher than the first because it is more difficult to remove an electron from a positively charged ion than from a neutral atom, the third higher than the second and so on, while the unqualified term is taken as the first

The two trends in the plot

- the periodicity of the graph is quite striking, with maxima at the noble gases, which have closed electron shells and very stable configurations, and minima at the alkali metals, whose low ionization enthalpies can be correlated with their high reactivity
- the first ionization enthalpy generally increases as we go across a period and decreases as we descend a group, and ionization enthalpy and atomic radius are closely related properties
- to understand these trends we have to consider two factors, the attraction of electrons towards the nucleus and the repulsion of electrons from each other

Shielding, and what it does to the trends

- the effective nuclear charge experienced by a valence electron will be less than the actual charge on the nucleus because of shielding or screening by the intervening core electrons, for example the 2s electron in lithium is shielded by the inner core of 1s electrons so it experiences a net positive charge less than the actual +3, and shielding is effective when the orbitals in the inner shells are completely filled, as in the alkali metals
- moving from lithium to fluorine across the second period, successive electrons are added to orbitals in the same principal quantum level and the shielding does not increase enough to compensate for the increased attraction, so increasing nuclear charge outweighs the shielding, the outermost electrons are held more and more tightly and the ionization enthalpy increases across a period
- going down a group the outermost electron is increasingly farther from the nucleus and there is increased shielding by the electrons in the inner levels, so here the increase in shielding outweighs the increasing nuclear charge and removal of the outermost electron requires less energy

Boron below beryllium, and oxygen below nitrogen

- the first ionization enthalpy of boron is slightly less than that of beryllium even though boron has the greater nuclear charge, because within the same principal quantum level an s-electron is attracted to the nucleus more than a p-electron, the electron removed from beryllium being an s-electron and that removed from boron a p-electron, and since the penetration of a 2s-electron is greater, the 2p electron of boron is more shielded and easier to remove
- another anomaly is the smaller first ionization enthalpy of oxygen compared to nitrogen, which arises because in nitrogen the three 2p-electrons reside in different atomic orbitals by Hund's rule, whereas in oxygen two of the four 2p-electrons must occupy the same 2p-orbital, increasing the electron-electron repulsion, so it is easier to remove the fourth 2p-electron from oxygen than one of the three from nitrogen

Problem 3.6, and electron gain enthalpy

- the question gives first ionization enthalpies for Na, Mg and Si of 496, 737 and 786 kJ mol-1 and asks whether the value for Al will be closer to 575 or 760 kJ mol-1, and the answer is closer to 575, because the value for Al should be lower than that of Mg due to effective shielding of the 3p electron from the nucleus by the 3s-electrons
- when an electron is added to a neutral gaseous atom to convert it into a negative ion, the enthalpy change accompanying the process is defined as the electron gain enthalpy, which provides a measure of the ease with which an atom adds an electron to form an anion
- depending on the element the process can be either endothermic or exothermic, and for many elements energy is released when an electron is added so the electron gain enthalpy is negative, the group 17 elements, the halogens, having very high negative values

Why the halogens and the noble gases sit at the extremes

- the halogens have very high negative electron gain enthalpies because they can attain a stable noble gas electronic configuration by picking up an electron
- the noble gases have large positive electron gain enthalpies, because the electron has to enter the next higher principal quantum level, leading to a very unstable electronic configuration

Table 3.H Electron Gain Enthalpies* / (kJ mol–1) of Some Main Group Elements

Group 1	ΔegH	Group 16	ΔegH	Group 17	ΔegH	Group 0	ΔegH
H	– 73					He	+ 48
Li	– 60	O	– 141	F	– 328	Ne	+ 116
Na	– 53	S	– 200	Cl	– 349	Ar	+ 96
K	– 48	Se	– 195	Br	– 325	Kr	+ 96
Rb	– 47	Te	– 190	I	– 295	Xe	+ 77
Cs	– 46	Po	– 174	At	– 270	Rn	+ 68

Arrows drawn on the table itself, one set across a period and one set down a group

Atomic radius falling across and rising down

Ionization enthalpy, electron gain enthalpy and electronegativity all rising across and falling down

Figure 3.H The periodic trends of elements in the periodic table

What electronegativity is, and is not

- electronegativity is the ability to attract electrons, and it is not constant for a given element but varies depending on the element to which it is bound
- though it is not a measurable quantity, it does provide a means of predicting the nature of the force that holds a pair of atoms together

The trends, and why they run that way

- electronegativity generally increases across a period from left to right, say from lithium to fluorine, and decreases down a group, say from fluorine to astatine
- the attraction between the outer or valence electrons and the nucleus increases as the atomic radius decreases across a period, so the electronegativity also increases, and on the same account the values decrease with the increase in atomic radii down a group, the trend being similar to that of ionization enthalpy

Table 3.I Electronegativity Values (on Pauling scale) Across the Periods

Atom (Period II)	Li	Be	B	C	N	O	F
Electronegativity	1.0	1.5	2.0	2.5	3.0	3.5	4.0
Atom (Period III)	Na	Mg	Al	Si	P	S	Cl
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0

Table 3.J Electronegativity Values (on Pauling scale) Down a Family

Atom (Group I)	Electronegativity Value	Atom (Group 17)	Electronegativity Value
Li	1.0	F	4.0
Na	0.9	Cl	3.0
K	0.8	Br	2.8
Rb	0.8	I	2.5
Cs	0.7	At	2.2

Electronegativity and metallic character

- non-metallic elements have a strong tendency to gain electrons, so electronegativity is directly related to the non-metallic properties of elements and inversely related to their metallic properties
- thus the increase in electronegativity across a period is accompanied by an increase in non-metallic properties, or a decrease in metallic ones, and the decrease down a group is accompanied by a decrease in non-metallic properties, or an increase in metallic ones

3.7.2 Periodic trends in chemical properties

Periodicity of valence or oxidation states

- most trends in chemical properties, such as diagonal relationships, the inert pair effect and the effects of lanthanoid contraction, are dealt with along with each group in later units, and what is studied here is the periodicity of the valence state and the anomalous properties of the second period elements from lithium to fluorine
- valence is the most characteristic property of the elements and can be understood in terms of their electronic configurations, being usually, though not necessarily, equal to the number of electrons in the outermost orbitals or to eight minus that number
- the term oxidation state is nowadays frequently used for valence, and the oxidation state of an element in a particular compound can be defined as the charge acquired by its atom on the basis of electronegative consideration from the other atoms in the molecule
- in OF2 and Na2O the order of electronegativity is F above O above Na, so each fluorine shares one electron with oxygen and is given oxidation state -1, oxygen sharing two electrons with the two fluorines and so showing +2, while in Na2O oxygen accepts two electrons, one from each sodium, and shows -2, sodium losing one electron and being given +1

Problem 3.8

- the question asks, using the Periodic Table, to predict the formulas of compounds which might be formed by silicon and bromine, and by aluminium and sulphur
- silicon is a group 14 element with a valence of 4 and bromine belongs to the halogen family with a valence of 1, so the formula would be SiBr4
- aluminium belongs to group 13 with a valence of 3 and sulphur to group 16 with a valence of 2, so the formula would be Al2S3

Table 3.K

Group	1	2	13	14	15	16	17	18
Number of valence electron	1	2	3	4	5	6	7	8
Valence	1	2	3	4	3,5	2,6	1,7	0,8

What the formulas show

- the hydrides and oxides an element forms follow its group, so the formula of the hydride runs LiH, CaH2, B2H6, CH4, NH3, H2O and HF across the periods, and the formula of the oxide runs Li2O, MgO, B2O3, CO2, N2O3 and N2O5, SO3 and Cl2O7
- reading down each column shows the same valence repeating, which is what makes valence a periodic property

Table 3.C Periodic Trends in Valence of Elements as shown by the Formulas of Their Compounds

Group	1	2	13	14	15	16	17
Formula of hydride	LiH, NaH, KH	CaH2	B2H6, AlH3	CH4, SiH4, GeH4, SnH4	NH3, PH3, AsH3	H2O, H2S, H2Se, H2Te	HF, HCl, HBr, HI
Formula of oxide	Li2O, Na2O, K2O	MgO, CaO, SrO, BaO	B2O3, Al2O3, Ga2O3, In2O3	CO2, SiO2, GeO2, SnO2, PbO2	N2O3, N2O5, P4O6, P4O10, As2O3, As2O5, Sb2O3, Sb2O5, Bi2O3	SO3, SeO3, TeO3	Cl2 O7

Problem 3.9, and the anomalous second period

- the question asks whether the oxidation state and covalency of Al in the chloro-aqua complex are the same, and the answer is no, the oxidation state of Al being +3 and the covalency 6
- the second element of a period resembles the second element of the following group, so lithium resembles magnesium and beryllium resembles aluminium, a similarity commonly referred to as a diagonal relationship in the periodic properties

3.7.3 Periodic trends and chemical reactivity

Why reactivity is high at both ends of a period

- all chemical and physical properties are a manifestation of the electronic configuration of elements, and since atomic and ionic radii generally decrease across a period, the ionization enthalpies generally increase and the electron gain enthalpies become more negative
- so the ionization enthalpy of the extreme left element in a period is the least and the electron gain enthalpy of the element on the extreme right is the highest negative, noting that the noble gases with completely filled shells have rather positive electron gain enthalpy values, which results in high chemical reactivity at the two extremes and the lowest in the centre
- the maximum chemical reactivity at the extreme left, among the alkali metals, is exhibited by the loss of an electron leading to a cation, and at the extreme right, among the halogens, by the gain of an electron forming an anion, which relates directly to metallic and non-metallic character, the metallic character being highest at the extreme left and decreasing while the non-metallic character increases from left to right

Reaction with oxygen, and Problem 3.10

- elements on the two extremes of a period easily combine with oxygen to form oxides, the normal oxide formed by the element on the extreme left being the most basic, such as Na_2O, and that formed by the element on the extreme right being the most acidic, such as Cl_2O_7, while the oxides of elements in the centre are amphoteric, such as Al_2O_3 and As_2O_3, or neutral, such as CO, NO and N_2O, amphoteric oxides behaving as acidic with bases and basic with acids, and neutral ones having no acidic or basic properties
- Problem 3.10 asks to show by a chemical reaction with water that Na_2O is a basic oxide and Cl_2O_7 an acidic one, Na_2O with water forming a strong base, 2NaOH, and Cl_2O_7 with water forming a strong acid, $2HClO_4$, their nature being qualitatively testable with litmus paper
- among the transition metals of the 3d series the change in atomic radii across the period is much smaller than for the representative elements, and still smaller among the inner-transition metals of the 4f series, while their ionization enthalpies are intermediate between those of the s- and p-blocks, so they are less electropositive than the group 1 and 2 metals

Down a group

- in a group, the increase in atomic and ionic radii with increase in atomic number generally results in a gradual decrease in ionization enthalpies and a regular decrease in electron gain enthalpies for the main group elements, with an exception in some third period elements
- so the metallic character increases down the group and the non-metallic character decreases, a trend that can be related with their reducing and oxidizing property
- in the case of transition elements, however, a reverse trend is observed, which can be explained in terms of atomic size and ionization enthalpy

Summary

How the table is built

- Mendeleev's Periodic Table was based on atomic masses, whereas the Modern Periodic Table arranges the elements in the order of their atomic numbers in seven horizontal rows, the periods, and eighteen vertical columns, the groups or families
- atomic numbers in a period are consecutive whereas in a group they increase in a pattern, and elements of the same group have similar valence shell electronic configuration and therefore exhibit similar chemical properties, while the elements of the same period have incrementally increasing numbers of electrons from left to right and therefore different valencies
- four types of elements can be recognized on the basis of their electronic configurations, the s-block, p-block, d-block and f-block elements, while hydrogen with one electron in the 1s orbital occupies a unique position

What the trends are

- metals comprise more than seventy eight per cent of the known elements, non-metals, which are located at the top of the periodic table, are less than twenty in number, and elements which lie at the border line between them, such as Si, Ge and As, are called metalloids or semi-metals
- metallic character increases with increasing atomic number in a group whereas it decreases from left to right in a period, and the physical and chemical properties of elements vary periodically with their atomic numbers
- periodic trends are observed in atomic sizes, ionization enthalpies, electron gain enthalpies, electronegativity and valence, the atomic radii decreasing from left to right in a period and increasing with atomic number in a group, while ionization enthalpies run the other way

Exercises

Exercises 3.1 to 3.9

- 3.1 What is the basic theme of organisation in the periodic table?
- 3.2 Which important property did Mendeleev use to classify the elements in his periodic table and did he stick to that?
- 3.3 What is the basic difference in approach between the Mendeleev's Periodic Law and the Modern Periodic Law?
- 3.4 On the basis of quantum numbers, justify that the sixth period of the periodic table should have 32 elements.
- 3.5 In terms of period and group where would you locate the element with Z =114?
- 3.6 Write the atomic number of the element present in the third period and seventeenth group of the periodic table.
- 3.7 Which element do you think would have been named by (i) Lawrence Berkeley Laboratory (ii) Seaborg's group?
- 3.8 Why do elements in the same group have similar physical and chemical properties?
- 3.9 What does atomic radius and ionic radius really mean to you?

Exercises 3.10 to 3.16

- 3.10 How do atomic radius vary in a period and in a group? How do you explain the variation?
- 3.11 What do you understand by isoelectronic species? Name a species that will be isoelectronic with each of the following atoms or ions. (i) F– (ii) Ar (iii) Mg2+ (iv) Rb+
- 3.12 Consider the following species: N3–, O2–, F–, Na+, Mg2+ and Al3+. (a) What is common in them? (b) Arrange them in the order of increasing ionic radii.
- 3.13 Explain why cation are smaller and anions larger in radii than their parent atoms?
- 3.14 What is the significance of the terms isolated gaseous atom and ground state while defining the ionization enthalpy and electron gain enthalpy? Hint : Requirements for comparison purposes.
- 3.15 Energy of an electron in the ground state of the hydrogen atom is –2.18×10–18J. Calculate the ionization enthalpy of atomic hydrogen in terms of J mol–1. Hint: Apply the idea of mole concept to derive the answer.
- 3.16 Among the second period elements the actual ionization enthalpies are in the order Li < B < Be < C < O < N < F < Ne. Explain why (i) Be has higher Δi H than B (ii) O has lower Δi H than N and F?

Exercises, continued

Exercises 3.17 to 3.24

- 3.17 How would you explain the fact that the first ionization enthalpy of sodium is lower than that of magnesium but its second ionization enthalpy is higher than that of magnesium?
- 3.18 What are the various factors due to which the ionization enthalpy of the main group elements tends to decrease down a group?
- 3.19 The first ionization enthalpy values (in kJ mol–1) of group 13 elements are B 801, Al 577, Ga 579, In 558, Tl 589. How would you explain this deviation from the general trend ?
- 3.20 Which of the following pairs of elements would have a more negative electron gain enthalpy? (i) O or F (ii) F or Cl
- 3.21 Would you expect the second electron gain enthalpy of O as positive, more negative or less negative than the first? Justify your answer.
- 3.22 What is the basic difference between the terms electron gain enthalpy and electronegativity?
- 3.23 How would you react to the statement that the electronegativity of N on Pauling scale is 3.0 in all the nitrogen compounds?
- 3.24 Describe the theory associated with the radius of an atom as it (a) gains an electron (b) loses an electron

Exercises 3.25 to 3.30

- 3.25 Would you expect the first ionization enthalpies for two isotopes of the same element to be the same or different? Justify your answer.
- 3.26 What are the major differences between metals and non-metals?
- 3.27 Use the periodic table to answer the following questions. (a) Identify an element with five electrons in the outer subshell. (b) Identify an element that would tend to lose two electrons. (c) Identify an element that would tend to gain two electrons. (d) Identify the group having metal, non-metal, liquid as well as gas at the room temperature.
- 3.28 The increasing order of reactivity among group 1 elements is Li < Na < K < Rb <Cs whereas that among group 17 elements is F > Cl > Br > I. Explain.
- 3.29 Write the general outer electronic configuration of s-, p-, d- and f- block elements.
- 3.30 Assign the position of the element having outer electronic configuration (i) ns2np4 for n=3 (ii) (n-1)d2ns2 for n=4, and (iii) (n-2) f 7 (n-1)d1ns2 for n=6, in the periodic table.

Exercises, continued

Exercise 3.31

- 3.31 The first and second ionization enthalpies and the electron gain enthalpy, all in kJ mol–1, of a few elements are given: element I at 520, 7300, –60; II at 419, 3051, –48; III at 1681, 3374, –328; IV at 1008, 1846, –295; V at 2372, 5251, +48; VI at 738, 1451, –40.
- 3.31, continued: Which of the above elements is likely to be (a) the least reactive element. (b) the most reactive metal. (c) the most reactive non-metal. (d) the least reactive non-metal. (e) the metal which can form a stable binary halide of the formula MX2 (X=halogen). (f) the metal which can form a predominantly stable covalent halide of the formula MX (X=halogen)?

Exercises 3.32 to 3.34

- 3.32 Predict the formulas of the stable binary compounds that would be formed by the combination of the following pairs of elements. (a) Lithium and oxygen (b) Magnesium and nitrogen (c) Aluminium and iodine (d) Silicon and oxygen (e) Phosphorus and fluorine (f) Element 71 and fluorine
- 3.33 In the modern periodic table, the period indicates the value of : (a) atomic number (b) atomic mass (c) principal quantum number (d) azimuthal quantum number.
- 3.34 Which of the following statements related to the modern periodic table is incorrect? (a) The p-block has 6 columns, because a maximum of 6 electrons can occupy all the orbitals in a p-shell. (b) The d-block has 8 columns, because a maximum of 8 electrons can occupy all the orbitals in a d-subshell. (c) Each block contains a number of columns equal to the number of electrons that can occupy that subshell. (d) The block indicates value of azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration.

Exercises, concluded

Exercises 3.35 to 3.37

- 3.35 Anything that influences the valence electrons will affect the chemistry of the element. Which one of the following factors does not affect the valence shell? (a) Valence principal quantum number (n) (b) Nuclear charge (Z) (c) Nuclear mass (d) Number of core electrons.
- 3.36 The size of isoelectronic species, F–, Ne and Na+, is affected by (a) nuclear charge (Z) (b) valence principal quantum number (n) (c) electron-electron interaction in the outer orbitals (d) none of the factors because their size is the same.
- 3.37 Which one of the following statements is incorrect in relation to ionization enthalpy? (a) Ionization enthalpy increases for each successive electron. (b) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration. (c) End of valence electrons is marked by a big jump in ionization enthalpy. (d) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Exercises 3.38 to 3.40

- 3.38 Considering the elements B, Al, Mg, and K, the correct order of their metallic character is : (a) B > Al > Mg > K (b) Al > Mg > B > K (c) Mg > Al > K > B (d) K > Mg > Al > B
- 3.39 Considering the elements B, C, N, F, and Si, the correct order of their non-metallic character is : (a) B > C > Si > N > F (b) Si > C > B > N > F (c) F > N > C > B > Si (d) F > N > C > Si > B
- 3.40 Considering the elements F, Cl, O and N, the correct order of their chemical reactivity in terms of oxidizing property is : (a) F > Cl > O > N (b) F > O > Cl > N (c) Cl > F > O > N (d) O > F > N > Cl